

Day 1

Gain Curve: Filename gcXX

Going slow from 800 volts in user wait. 30 second dwell for the gain curve

During MCP gain curve, observed no outgassing

1500 V on MCP, first counts 0.2 Hz, GC08 filename. Current 0.751

Turned science chamber ion gauge off.

Saving a true dark counts file as DC01 with ion gauge off in science chamber.

1675V on MCP: WE SEE THE BEAM!!!

1700V we also observe a patch at the bottom of the MCP

11:12 turned output on on HVPS. 250 V

500V

FL01 = First light, 0 V on deflector plates. 7.7 Hz

FL02 = First light, 250 V on deflector plates

FL03

FL11 11:58AM - GRAPHENE! 1 keV p+

After FL12 we moved the rotational stage back to Home and closed the beamline GV. Now going to dwell for 5 minutes for another dark count

We believe that the structure seen at 6 o'clock is the ion gauge

15:10 Resumed experiment

Day 2 April 24

In the lab at 8AM sharp, Lengying and I started RGA and ion source startup procedures

Ramped up MCP with the beamline GV and science chamber GV open.

Forgot to turn on Helmholtz coil

Turned on Helmholtz coil and monitored counts. Verified ~30 Hz

Turned off science chamber ion gauge at 09:13. Last reading was $1.2\text{E-}7$ Torr

Plan for today:

- Dark Count test
- Carbon Foil Electron Emission 1 keV p+
- Graphene Foil Electron Emission 1 keV p+
- Carbon Foil Electron Emission 4 keV p+
- Graphene Foil Electron Emission 4 keV p+
- Carbon Foil Steering Test 4 keV p+

09:35

Closed Beamline GV and increasing pressure to $1\text{E-}5$ to get better counts

Secondary electrons: We think we see a cluster maybe? But it's hard to tell with statistics

Ramping voltage down on the deflector plates until we get to 0. Then will close the beamline GV, switch to graphene, and test at 4, 5, 6 V.

Graphene we are not seeing any structure.

Will go 6, 5, 4 V on graphene at 4 keV p+ beam. Then 6, 5, 4 V on amorphous carbon

Science Chamber Opening Tests: Be careful about cheerios. Use laser to ensure that all foils are lined up

Lengying thinks that maybe our foil floating process could be introducing water into the graphene

11:07

Ok we are switching test campaigns now. We will keep the 4 keV p+ beam but now attempt to steer the charge state populations with better statistics. Disconnected the small blue power supply and connecting the two HVPS. Will ramp to 2000 V

Jamie suggests checking the ratio of the distance of the steered dot to the 750 V to the ratio of 2500, 3000, and 3500V to the 4 keV beam. (Do this during data analysis)

YAYY we see good separation with the deflector plates! So at a 1 keV beam it's just statistics.

UP NEXT: Shine a 16 keV beam through the graphene to try to punch through the graphene

11:30

Turning on Matsusada to float the beam up to 16 keV. Going up in 1 keV steps. Hypothesis to see what the stopping power of graphene is. Will record actual beamline voltage. Won't save each step but just do sanity checks to see when we start seeing stuff on the MCP.

5006.20 on Total Beamline Voltage. Opening Beamline GV. About 3 Hz

Closed Beamline GV. Total Beamline Voltage: 6005.87 V. Ramping down Deflector Plates since we're just trying to see if anything is coming through the graphene at all. Still about 3 Hz

Closed GV. Increased Total Beamline Voltage to 7006.28 V. Opened Beamline GV. Still 2-3 Hz. Completed rampdown of HVPS and powered them off.

Closed GV. Total Beamline Voltage 8006.24 V. 5 Hz now. There is no deadtime so we're not saturating the MCP or anything. 4.88 Hz. Saving this file as G01. Ion Source Pressure 2E-5 Torr

Closed GV. Total Beamline Voltage 9004.63 V. Open Beamline GV. 5.99 Hz.

Closed GV. Total Beamline Voltage 10006.25 . Open Beamline GV. ~ 4 Hz

Closed GV. Total Beamline Voltage 11006.25 V. Open Beamline GV. 7.54 Hz

Closed GV. Total Beamline Voltage 12004.84 V. Open Beamline GV. 8.59 Hz

Closed GV. Total Beamline Voltage 13002.27 V. Open Beamline GV. 9.29 Hz

Closed GV. Total Beamline Voltage 14004.52 V. Open Beamline GV. 10.28 Hz

Closed GV. Total Beamline Voltage 15002.03 V. Open Beamline GV. Pressure is still 2E-5 Torr in the Ion Source. We see three hotspots around the perimeter of the MCP. Saving this one as G03. 11.49 Hz.

Last test. Closed Beamline GV. Total Beamline Voltage 16002.88. Open Beamline GV. 12.33 Hz. Saved as G04.

Shutting beamline down at 12:01 PM.

Look into the electrostatic properties of graphene. Maybe it's charging up and acting as an insulator?

Dwells from this Matsusada test campaign were 120 seconds.